Asset Performance:

Ene 10 Demand side management (DSM) capabilities for electricity

0 4

Credits No Minimum Standard

Aim

To reduce carbon emissions associated with grid supply electricity by enabling electricity demand profiles to better match the availability of renewable electricity generation sources.

Question

1. Does the asset have any electric storage capacity?
2. Is cogeneration optimised to coordinate with local renewable energy generation and local energy demand profiles?
3. Are electric smart appliances or electric domestic hot water (DHW) subject to DSM control?
4. Is electric heating subject to DSM control?
5. Is electric cooling subject to DSM control?
6. Does the electric vehicle charging, or other charging loads, include any grid balancing?
7. Does the HVAC system have run time management?

Credits	Question	Select a single answer option	Points awarded	Points available
*	1	Yes	1	1
		No	0	1
*	2	No cogeneration	Filtered	0
		Yes	1	1
		No	0	1
*	3	Yes	1	1
		No	0	1
*	4	No electric heating	Filtered	0
		Yes	1	1
		No	0	1

Credits	Question	Select a single answer option	Points awarded	Points available
*	5	No electric cooling	Filtered	0
		Yes	1	1
		No	0	1
*	6	No electric vehicle or other charging loads	Filtered	0
		No	0	2
		On-way grid balancing controlled charging	1	2
		Two-way grid balancing controlled charging and electric vehicle to grid	2	2
*	7	No HVAC system	Filtered	0
		Yes	1	1
		No	0	1

* Refer to Methodology

Methodology

Credit allocation

Credits are based on the percentage of points achieved against the number of points available as follows;

Table 21: Allocation of Credits

Percentage of available points achieved	Credits
≥ 25%	1
≥ 50%	2
≥ 75%	3
100%	4

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the ‘BREEAM evidential requirements’ section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.

Criteria	Evidence requirement
All	Extract of relevant O&M manuals OR a copy of manufacturer information with link to the project assessed.
All	Visual inspection and verification.

Definitions

Charging loads:

Building related charging loads, such as charging rooms, represent a significant way of making electricity demand and grid supply match.

Demand side management (DSM):

Demand-side management refers to adopting measures to improve consumption efficiency on the demand side by responding to electricity generation. Changing consumption patterns reduces overall electricity consumption and therefore demand, but meets the same consumption function. This could include but not limited to;

- A reduction in electrical consumption of traditional appliances
- Increase in time-of-use appliances and electronic devices
- Plug-in electric vehicles grid balancing
- Local renewable electricity generation
- Control devices and platforms
- Electricity storage technologies.

Grid balancing:

Using this capability to ensure that electricity input to the grid matches the electricity demand. This results in a reduction in carbon emissions compared to conventional grid balancing which involves ramping up existing fossil fuelled power plants.

Run time management:

This is a type of control that limits the hours that a piece of equipment is able to run. This would involve individual setting following a predefined time schedule including fixed preconditioning phases.